

GENESIS based implanted neuro-stimulation device to treat chronic migraine

A global medical-device company, St. Jude Medical Inc., has come up with a GENESIS based implanted neuro-stimulation device for patients with intractable chronic migraine, which is one of the most difficult-to-treat headache disorders. This system is used for the peripheral nerve stimulation (PNS) of the occipital nerves for management of pain and disability associated with the migraine. The software developed in GENESIS is encoded into a neuro-stimulator and is connected to small electrical lead/leads placed under the skin. It provides mild electrical stimulation pulses to the occipital nerves that are located just beneath the skin at the back of the head. A study on 157 patients, with each patient suffering from headache for about 26 days per month, reported that 65 per cent of patients reported excellent or good pain relief.

First, unlike in dedicated simulations, we can modify the existing code for including new components. This dramatically boosts-up the speed of modelling, which is the time-consuming part in most of simulations.

Second, model components can be shared between different simulations. If we are able to use the previously-developed-and-tested components, it can easily boost-up the speed of new simulations.

Third, general simulations can serve as a means of communication between different laboratories interested in the same system. GENESIS allows researchers to test the modelling results of other researchers working in the same area.

Beyond this, there are a lot of technical advantages, such as assurance on credibility of software core of simulations, accessibility to parallel computers, new integration techniques or graphics and some form of standardisation for model descriptions.

Design features

Let us take a look at the key design features of this tool.

Building-block approach. Simulations and the graphical user interface (GUI) in GENESIS are based on the building-block (modular) approach. Each building block performs well-defined functions (that is, receiving input, performing computations, producing output) and communicates with each

other in a standardised way. The advantage of using this approach for neural simulations is that, these systems typically consist of discrete components that interact with each other in quite stereotypical ways, and more simulations use similar neural components, display routines, numerical-integration routines, etc.

The user can choose any combination of these blocks and extend the tool for a new application by adding new blocks with the help of an object-oriented design. Individual modules or linked assemblies of modules can easily be replicated.

Flexible interaction. GENESIS allows the user to input commands through Script files, GUI or using GENESIS command shell.

Script language interpreter (SLI) is the underlying level of GENESIS user interface. This interpreter has an extensive set of commands related to the building, monitoring and controlling of simulations. GENESIS simulation objects and graphical objects are linked together using the scripting language. The interpreter can read SLI commands either interactively from the keyboard (allowing interactive debugging, inspection and control of the simulation) or from files containing simulation scripts.

GENESIS has a smart GUI called X-Windows Output and Display Utility for Simulations (XODUS). Users can easily develop simulations and monitor these with the

help of XODUS. The set of graphical modules present in this interface is exactly the same as that of computational modules from the user's perspective. Just like in the case of computational modules, the user can choose the manner in which data can be entered or displayed.

Also, graphical modules ensure full power of SLI, with the provision of calling functions from the script language.

Independent of the choice of device. Large-scale neural simulations involve a large number of groups working on the same project. For making simulations available to a large number of users, the design of the software is made to run under UNIX and X-Windows. It is tested on a large number of machine architectures. The implementation of PGENESIS has also expanded the scope of issues that GENESIS can address.

A powerful neural simulator

Over the years, GENESIS has emerged as a powerful large-scale neural simulator. Although GENESIS 2.3 was intended to be a final release of the GENESIS 2 series, it has continued to evolve with modifications needed to efficiently implement synaptic plasticity in large networks and many contributions from users. This has led to a new November 2014 GENESIS 2.4 release, which is incorporated in this month's EFY Plus DVD. In order to address issues like difficulty in adding more modern Java based graphical interfaces, alternate script parsers and interfacing via the World Wide Web, GENESIS 3 is also being developed in parallel.

Hope you have tried the iqr neural-simulation software that we had introduced in last month's EFY Plus DVD. This month why not give GENESIS a try? ●

The author worked at EFY till recently